

=== NETWORK PROJECT CHANGELOG ===

Start Date: 2026-07-08

--- 2026-07-07 ---

11:20 – Created initial topology in Packet Tracer, with 2 Routers, 2 multilayer switches, 3 access layer switches, 6 PCs and 2 Servers.

--- 2026-07-08 ---

8:17 – Configured the hostnames of each device.

8:20 – Assigned each interface connected to an access layer switch to its vlan.

8:29 – On the Multilayer switches, I added SVIs to act as the default gateway to each vlan.

8:37 - Configured IP on PC1, tested ping to gateway. FAILED.

8:40 - Root cause: Missing trunk between Access-SW1 and Dist-SW1.

8:46 - Configured trunk on both ends (interface trunk mode on, allowed VLANs).

8:48 - Retested: ping to gateway SUCCESS.

8:54 – Configured trunks between my access layer switches and distribution layer switches.

9:03 – Configured DHCP excluded addresses on Access-SW3, the IP- addresses of default gateways were excluded to prevent IP conflict.

9:06 – Configured a DHCP pool for vlan 10 named POOL_VLAN10, I set the network, default-router and DNS-server.

9:36 – Fixed my mistake of creating the same SVIs on both distribution switches with the same IP addresses. Instead, I use HSRP to create virtual IPs which they will both have.

10:02 – On both distribution switches I have configured them as DHCP relay agents for vlan 10.

10:08 – Renewed IP on PC2, SUCCESS.

10:12 – Created POOL_VLAN20 and POOL_VLAN30 to provide IP addresses to vlan 20 and 30.

10:17 – Used SVIs on DSW1 and DSW2 to relay IP addresses.

10:20 – On PC4 tried to renew IP-address. FAILED.

11:00 - Root cause: Broadcast storms caused by STP, all the ports in vlan 20 were forwarding ports.

11:20 – Manually configured root bridge as well as bridge ids of each bridge.

11:40 – Retested: IP was received but in the wrong subnet. FAILED.

12:00 – Root cause: DHCP configurations were applied to SW3 which has no IP to route to when a device needs an IP address.

12:34 - Removed DHCP configurations on SW3 and applied them to R2, allowing all devices regardless of the vlan or subnet they are in to be able to access the DHCP server.

--- 2026-07-09 ---

9:21 – Added IP addresses on DSW1, DSW2, R1 and R2 and enable routing on the multilayer switches.

9:33 – Removed the DHCP relay command to the old DHCP server and changed the IP address to point to R2.

9:36 – Retested: Checked if PC3 could renew it's IP. FAILED.

10:00 – Root Cause: No routes between the DHCP sever and the PCs.

10:20 – Added OSPF to all routers and distribution switches. Statically added an IP to PC2 and tested connectivity. SUCCESS.

10:25 - Retested: Checked if PC2 could renew its IP. FAILED.

11:30 – Root Cause: “Sevice dhcp” command wasn’t used, so the router couldn’t act like a DHCP server.

11:32 – Retested: Checked if PC2 can renew its IP. SUCCESS.

--- 2026-07-12 ---

8:37 – Applied the DHCP relay on vlan20 and 30 interfaces.

8:40 – Tested to see if endpoints in vlan 20 and 30 gained an ip. SUCCESS.

9:20 - Added serial connections between core routers and ISP router.

9:24 – Tested if I could ping the interfaces on the ISP router. SUCCESS.

9:26 – Added a loopback interface of 8.8.8.8 on ISP to imitate a DNS server for future pings.

9:30 – Added OSPF configurations on ISP router.

9:32 - Test connectivity between PCs and ISP by pinging 8.8.8.8 on PC1. SUCCESS.

9:40 – On R1 I applied NAT overload configurations. I set so/2/0 as the outside interface and go/1 and go/2 as inside interface.

9:44 – Tester: Checked Nat table to see if translations occurred. FAILED

9:46 – Root cause: No data has left R1's s0/2/0 interface.

9:49 – From PC3 I pinged 8.8.8.8 since that ip was outside my network.

9:51 – Tested: Checked NAT table again. SUCCESS.